Groups keep the amount - learner book

Draft • M01 • L-MATH-001-01-01 • tens and ones, 0-99

Use counters, groups of ten, paper and a pencil. Directions may be read aloud. You may speak, draw, point to objects, or ask an adult to write your explanation. A partner may be another learner or your teacher.

Our question

How can the groups change while the amount stays the same?

Say or show: one counter is our one-unit; one group of ten contains ten counters. Use the words one, ten, group, amount and equal.

Cycle A - Build and describe

Watch your teacher group fourteen counters. Name the grouped units. Explain what happened to the counters when a group was made.

A1 - Make groups

Start with sixteen counters. Make as many complete groups of ten as possible. Show the tens and the extra ones. Explain how you know no counters disappeared.

My model and explanation:

A2 - Listen and build

Your teacher will describe two collections. Build each one and label its tens and extra ones. Explain any zero position in your written numeral.

First collection:

Second collection:

A3 - Your partner has a clue

Keep your own card private. Ask your partner about the box whose clue you do not have. Build or draw the groups from the description; ask if your model agrees. Then swap roles. Show each complete collection with tens and extra ones.

Red box:

Blue box:

A4 - Show what you understand

Build or draw twenty-seven counters in complete tens and extra ones. Name the counted one-unit. Explain why making ten-groups keeps the amount.

My model and reason:

Cycle B - Exchange and explain

Watch your teacher unpack one ten from a collection of twenty-one counters. Ask: did any counters enter or leave the collection?

B1 - Unpack a ten

Build thirty-two with complete tens and extra ones. Unpack one ten. Show both stages. Explain why they have equal amounts.

Before: After:

My reason:

B2 - Test a claim

A learner says: “If I unpack one ten from four tens and three ones, I get three tens and three ones.”

Build or draw the change. Is the claim correct? Correct it and give a reason.

My correction and reason:

B3 - Reconstruct the other collection

Use Cycle B on your separate partner cards. Describe your collection without showing the card. Your partner reconstructs it, packs complete tens, and explains why the total stayed the same. Swap roles.

Green box in complete tens and extra ones:

Gold box in complete tens and extra ones:

Our explanation of the exchange:

B4 - Two ways, one amount

Show forty-six as complete tens and extra ones. Show it again after unpacking one ten. Explain to your partner why both models still show forty-six.

First model: Second model:

My explanation:

Think back

Show your partner one task you can explain. Name one task you want to try again. Your teacher will check the mathematics with you; confidence alone is not a result.

Groups keep the amount - workbook

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Practice only after the matching cycle. Use counters or drawings. An adult may read the directions and record your spoken explanation. Stop and ask for another demonstration when a group does not make sense.

After Cycle A

W1

Show twenty-eight with complete tens and extra ones. Label both units.

Model:

W2

Draw four tens and nine extra ones. Write the total numeral.

Model and numeral:

W3

Build seventy. Explain what its zero ones means.

Model and reason:

After Cycle B

W4

Build five tens and one extra one. Unpack one ten. Show both equal amounts.

Before: After:

W5

Start with two tens and fifteen ones. Pack complete tens. Name the complete tens and remaining ones.

Before: After:

W6

Someone says: “Ten ones can be exchanged for one one.” Correct the named unit and explain why it has the right value.

Correction and reason:

W7

Show thirty-eight in two ways: complete tens and extra ones, then after unpacking one ten. Label the units.

First: Second:

W8

Explain why the exchange in W5 changed the groups but kept the amount. Use a picture, objects or words another person can follow.

Explanation:

Numerals, names and sums

Grade 2 • draft • L-MATH-001-01-03 • PV2, 0-99

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV2-03-01

Write thirty-nine as a numeral and a sum of place values. Build or draw the same amount.

Model / numeral / sum:

P-PV2-03-02

Build sixty. Write its numeral and name. Explain the zero position.

Model / name / reason:

P-PV2-03-03

Someone says the 6 in 68 means six single ones. Test the claim with groups. Explain your conclusion.

Model / conclusion / reason:

P-PV2-03-04

Read 82. Write its name and expanded form. Explain what the 8 contributes.

Name / sum / explanation:

Counts and equal steps

Grade 2 • draft • L-MATH-001-01-04 • PV2, 0-99

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV2-04-01

Give five terms by fives starting at 55. Include 55 as the first term. Explain the step.

Terms / step:

P-PV2-04-02

Count by ones from 58 through 62, including both endpoints. Show the groups just before and after 59 becomes 60. Explain the exchange after adding one.

Terms / before and after models / reason:

P-PV2-04-03

Give three terms by tens starting at 46. Include 46. Explain what changes and what stays the same.

Terms / explanation:

P-PV2-04-04

A person counts 25, 30, 36, 40, 45 and calls every step five. Test the claim, correct the sequence and explain.

Corrected terms / reason:

Compare amounts and explain

Grade 2 • draft • L-MATH-001-01-05 • PV2, 0-99

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV2-05-01

Put 19, 29 and 91 in increasing order. Show why your order works.

Order / model / reason:

P-PV2-05-02

Complete 57 __ 75 using <, > or =. Read your comparison and explain using the places.

Comparison / reason:

P-PV2-05-03

Complete 82 __ 78 using <, > or =. Explain using the first differing place.

Comparison / reason:

P-PV2-05-04

Compare 66 and 66. Someone says 90 is less than 49 because 0 is less than 9. Test both comparisons and explain.

Both comparisons / models / reasons: